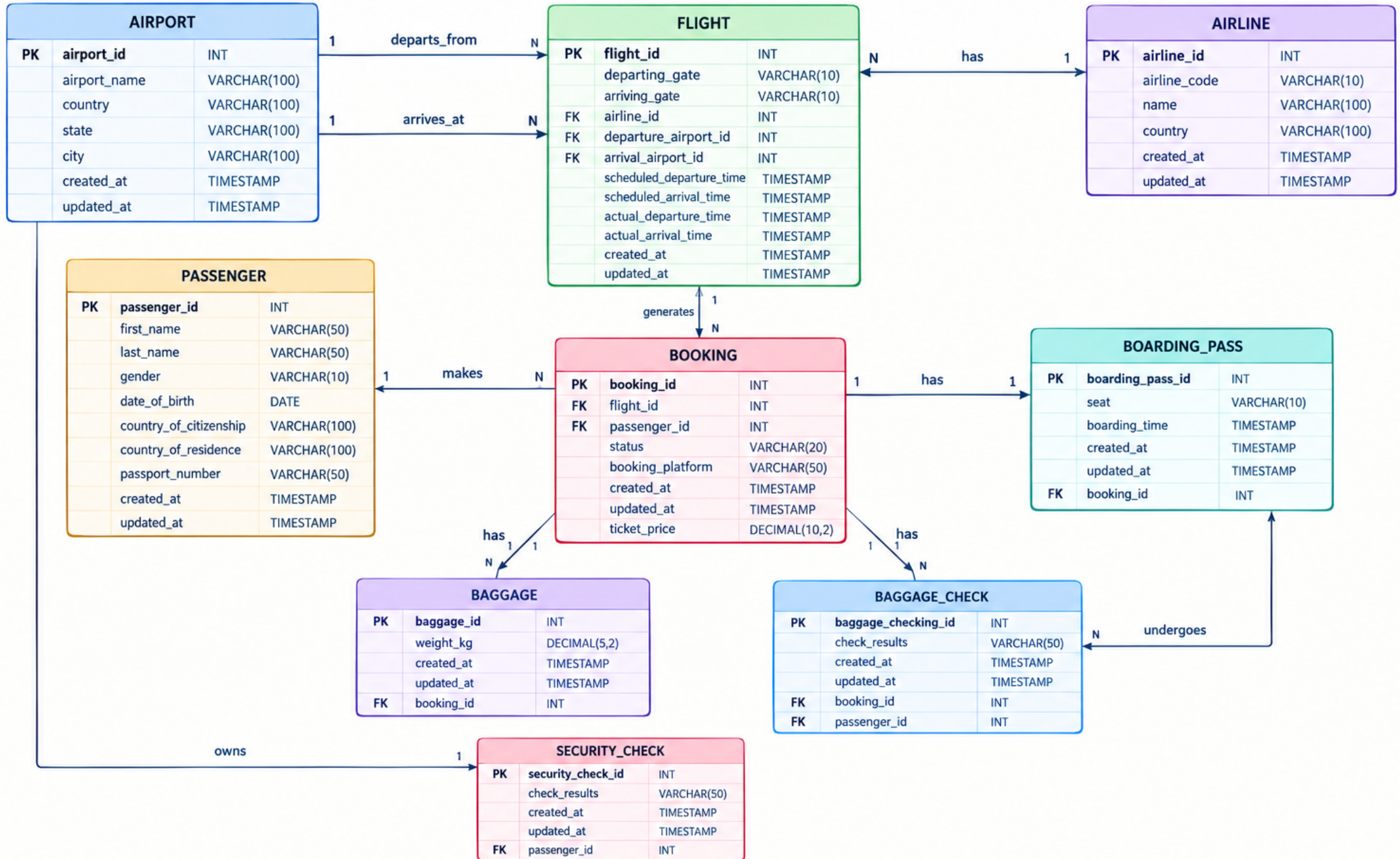


International Airport – ER Diagram

Lab 1: Entity-Relationship Diagram



Explanation

1. Airport

Stores airport_id, airport_name, country, state, city, created_at and updated_at. airport_id is the primary key.

2. Airline

Stores airline_id, airline_code, name, country, created_at and updated_at. One airline can have many flights (1:N).

3. Flight

Stores flight, gate, scheduled/actual time and timestamp information. airline_id, departure_airport_id and arrival_airport_id are foreign keys. A flight has one airline and one departure and arrival airport.

4. Booking

Stores booking_id, flight_id, passenger_id, status, platform, timestamps and ticket price. It connects passengers to flights. One flight and one passenger can each have many bookings.

5. Passenger

Stores passenger identity/profile information including name, gender, date of birth, citizenship, residence, passport number and timestamps.

6. Boarding Pass

Stores boarding_pass_id, seat, boarding_time, timestamps and booking_id. This diagram represents Booking → Boarding Pass as 1:1.

7. Baggage

Stores baggage_id, weight_kg, timestamps and booking_id. One booking can have many baggage records (1:N).

8. Baggage Check

Stores baggage checking id, check results, timestamps, booking_id and passenger_id. It is associated with both a booking and a passenger.

9. Security Check

Stores security_check_id, check results, timestamps and passenger_id. It is associated with a passenger.

Legend

PK = Primary Key. **FK** = Foreign Key. **1:N** = one-to-many. **1:1** = one-to-one.